3. DESIGN APPROACH

The R.E.P. (***Repetition***, ***Efficiency***, ***Precision***) Tracker provides real-time form analysis and vital monitoring to enhance weightlifting safety and effectiveness. Critical requirements driving the approach include sustained battery life for extended use, accurate motion tracking with less than 2% error, vital measurements with low error margins, clear video capture at 1080p, and operation in standard gym environments from 0^{o} to 40^{o} C with up to 85% humidity. Constraints encompass portability in a tote-around case, affordability, and compatibility with common gym equipment. Standards such as FCC Part 15 [1] for ensuring the device does not cause interference with other wireless devices in gyms guided by the design.

During the initial design phase for the R.E.P. Tracker, various solutions were evaluated. The final solution is a portable system that includes a wireless battery power subsystem, motion tracking subsystem, performance monitoring subsystem, user feedback subsystem, user interface subsystem, and camera subsystem. The sensors and interface allow the user to monitor form and vitals in real time, while vocal and visual feedback help the user adjust exercises and track progress. These components enhance weightlifting safety and effectiveness, boosting user confidence. The following subsection covers the overall design choices that were made during the development of the R.E.P Tracker and how each meet project requirements.

3.1 Design Options

Two design options were considered for the R.E.P Tracker: a corded system and a cordless system. The corded design offered constant power and reliable data transmission without interference, but it limited user mobility, posed safety risks from cables, and reduced portability. The cordless design, powered by rechargeable batteries and using wireless communication, provided full mobility, portability, and user convenience but required efficient power management and careful adherence to FCC Part 15 standards [1] to prevent interference. The team selected the cordless design because it best met the project's requirements for accuracy, usability, and safety in gym environments while maintaining performance within acceptable cost and power limits.

3.1.2 Design Option 1

The first design option for the R.E.P Tracker was a corded system that used direct power connections to supply continuous energy and ensure reliable, interference-free data transmission. This setup offered simplicity, reduced costs, and eliminated the need for batteries or wireless modules, making it a stable and consistent option for operation. However, the corded design severely limited user mobility, introduced potential safety hazards from tangled or hanging cables, and conflicted with the project’s requirement for portability and ease of use in a gym setting. While dependable in performance, this option was ultimately not suitable for a device intended to support unrestricted movement and convenience during workouts.

3.1.3 Design Option 2

The second design option for the R.E.P Tracker was a cordless system powered by rechargeable batteries, which introduced wireless data transfer to a central processing unit. This wireless communication was identified as the most challenging aspect of the design, as it required reliable, low-latency transmission while maintaining compliance with FCC Part 15 standards [1] to prevent interference with other gym devices. Despite this complexity, the cordless system offered significant advantages, including full mobility, portability, and a safer, more convenient user experience by eliminating cables that could obstruct movement or create hazards. By incorporating low-power sensors and efficient battery management, the system could provide accurate real-time motion tracking and vital monitoring with minimal downtime. While more complex and slightly higher in cost than a corded design, this option aligned best with the project's requirements for usability, performance, and safety in gym environments.

3.2. System Overview

At its core, the R.E.P. Tracker functions as a smart coaching system that tailors workouts based on user input and provides real-time feedback throughout the exercise session. As shown in Figure 3-1, the black box diagram illustrates the overall system flow. Powered primarily by a 12.8 VDC main battery—with supplemental batteries for individual sensors—the device collects and processes workout data internally. It then delivers visual and audio-based feedback and logs performance data for ongoing tracking and improvement.

Figure 3-1 shows the basic input output diagram for the R.E.P. Tracker.

Figure 3-1: The R.E.P. Tracker System at a Glance (Level 0)

Figure 3-2 shows the innerworkings of the R.E.P. Tracker, elaborating how the subsystems interact.

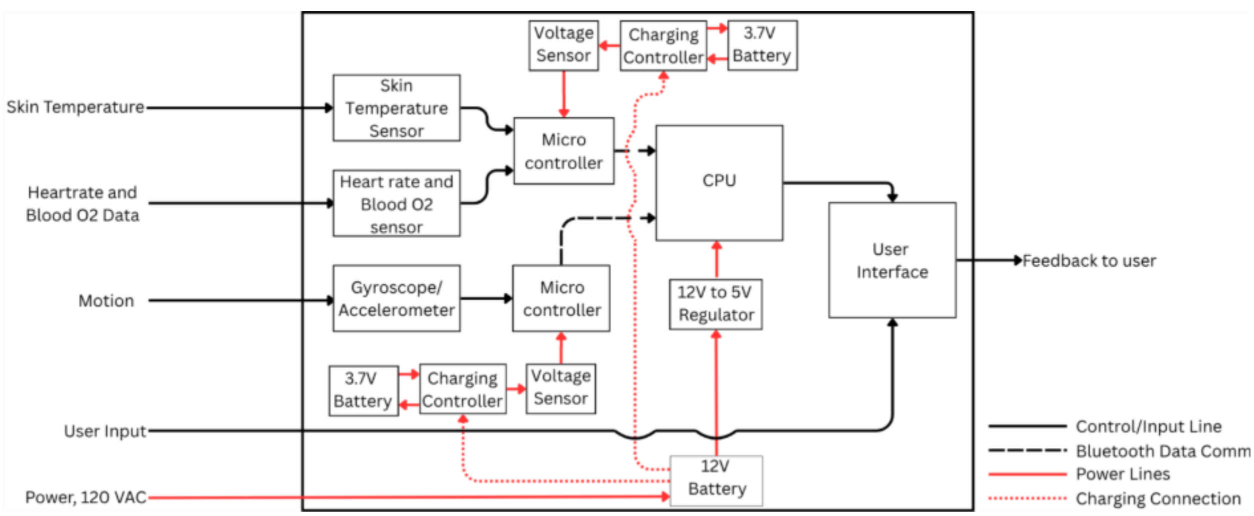

Figure 3-2: The R.E.P. Tracker Functionality (Level 1)

The R.E.P. Tracker is designed to act as a passive yet intelligent workout assistant, providing real-time feedback with minimal user input. Once the user enters their personal data and selects the desired workout through the User Interface (UI), the system takes over. The Motion Tracking subsystem monitors body movements to calculate repetitions and sets, while the Performance Monitoring subsystem displays vital signs on the UI as needed. The User Feedback system processes data from both Motion Tracking and Performance Monitoring to deliver real-time guidance and alerts, supported by the Camera Feed, which visually highlights any form or movement errors. All components are powered by a magnetic charging system that energizes the sensors and main microcontroller. Together, these subsystems work seamlessly to provide an efficient, accurate, and engaging workout experience.

3.2.1. Microcontroller

The central processor and microcontroller are the core systems of the R.E.P. Tracker project. All of the subsystems communicate and are connected via these controllers. Table 3-1 shows the options for possible Central Processors, and Table 3-2 shows the options for Sub Microcontrollers.

Table 3-1: Central Processor Options

Processor	**Ram**	**Bluetooth and Wi-Fi capabilities**	**Operating system**	**Price**
Requirements	4GB minimum	yes	Linux	<$100
Raspberry Pi 5 [2]	8GB	yes	Linux	$81.19
Arduino Uno [3]	2KB	no	Linux	$19.99
ESP32 [4]	512MB	yes	FreeRTOS	$10

Table 3-2: Sub Microcontroller Options

Processor	Ram	Bluetooth	Opporation System	Price
Requirements	250 KB minimum	yes	Linux	<$10
Raspberry Pi Pico 2w [5]	520 KB	yes	Linux	**$7**
Raspberry Pi Zero 2W [6]	512MB	yes	Linux	$23.91

Figure 3-3, seen below, shows the chosen central processor and the chosen microcontroller.

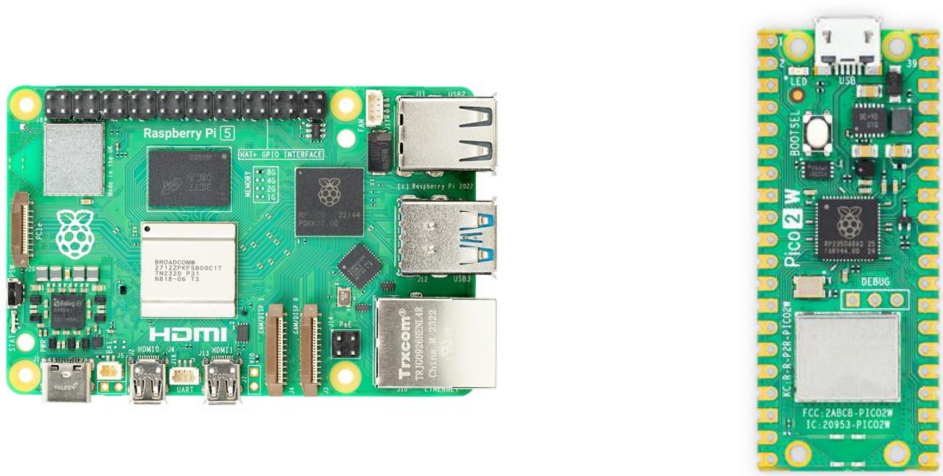

Figure 3-3: Raspberry Pi 5 (left) [2] and Raspberry Pico 2W (right) [5].

Both the Raspberry Pi 5 and Raspberry Pico 2W were chosen because the R.E.P. Tracker will communicate between modules via Bluetooth, of which they have sufficient Bluetooth capabilities. The central processor for the R.E.P. Tracker also requires a decent amount of data processing power capability; respectively, the 8GB of RAM provides enough processing to run the programs and receive all the data for the system. Both the Pi 5 and Pico 2W operate through the Linux OS, which is readily accessible, quickly learnable, and sufficiently compatible with each subsystem. The alternative microcontrollers for the REP Tracker include the Arduino Uno and the ESP32. While both are decent microcontrollers, the Arduino would not have as much processing power as the Pi 5. The Pi 5 is also easier to program and is more compatible with the programs and components used in this project. Another microcontroller that was considered for the project was a previous Raspberry Pi model like the Pi 4. These models were powerful enough to run the software; however, many of the previous models of Pi do not have the built-in Wi-Fi and Bluetooth that are essential for the REP Tracker.

3.3. Subsystems

The R.E.P. Tracker prototype consists of three primary subsystems: Motion Tracking, Power System, and User Interface. Performance Monitoring serves as a backup for initial testing. For the final system phase, Performance Monitoring, Camera Feed, and User Feedback will be fully integrated.

Below is a list of the subsystems for the R.E.P. Tracker and a brief description of each.

1) Motion Tracking: This subsystem processes raw motion data for monitoring and trending.
2) User Interface: The UI subsystem displays various metrics and real-time feedback from all other subsystems in an intelligible form to the exerciser. Additionally, the UI allows the user to edit and select customizable R.E.P. Tracker options.
3) Power System: This subsystem is a muti-stage battery network that allows the whole system to be powered cordlessly (not requiring a wall outlet) and readily rechargeable.
4) Performance Monitoring: The PM subsystem monitors and communicates vitals to the rest of the subsystems for tracking and trending.
5) User Feedback: This subsystem monitors key metrics from the MT, UI, and PM subsystems to give real-time feedback to the exercisers' workout form and to output auditorial/visual suggestions.
6) Camera Feed: The CF subsystem records and streams a quality, live video for the user’s benefit in monitoring and correcting exercise form.

3.3.1. Motion Tracking

Motion Tracking (MT) uses motion-based metrics acquired via an accelerometer, gyroscope, and/or magnetometer to receive real-time data during a user’s exercise "repetition". The acquired data is gathered by the Pico and then sent via the Pico’s Bluetooth communication protocols. The Pi 5 then processes this data using relevant logic. Finally, the readable data can be shared with any other subsystem (namely the User Interface) for tracking and trending purposes.

A module was chosen that includes all three of the motion sensor metrics - an accelerometer, a gyroscope, and a magnetometer. The rationale is that by using three separate motion modules, the data will have improved continuous accuracy. Congruently, calibration of the sensor module is steadier and more permanent with three sensors being utilized. This sensor module will connect to the Pico microcontroller via I2C protocols and GPIO pins. As seen below, Formulas (1) and (2) show the mathematics governing much of the MT processing logic, and Table 3-3 shows the options for the MT sensor module.

Mathematics of REP Tracker MT subsystem

$$\text{accel mag} = \sqrt{x^2 + y^2 + z^2 - |G(x,y,z)|^2} \tag{1}$$

$$\text{gyro average} = \frac{\omega_x + \omega_y + \omega_z}{3} \tag{2}$$

Table 3-3: Motion Tracking Module Options

Product	**Voltage Rating(V-DC)**	**Total Axes**	**Weight (oz)**	**Dimensions (in)**	**Cost (USD)**
Requirements	-	Error: < 2%	< 2	< 3 x 3 x 1	< 50
HiLetgo MPU-6050 [7]	3~5	6	0.63	3 x 2.5 x 0.5	~7
Adafruit BNO055 [8]	2.4~3.6	9	0.106	1.1 x 0.8 x 0.2	~35
Sparkfun IMU 15335 [9]	1.95~3.6	9	~0.106	1 x 0.7 x 0.2	~22

Figure 3-4 displays the motion sensing module chosen for the MT subsystem. This module was selected because it supplies 9-axes of accurate data through its pre-installed Fusion calibration hardware, and it has a minimum profile (size and weight).

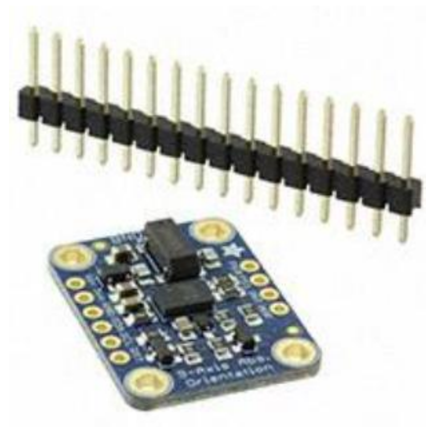

Figure 3-4: Adafruit 9-DOF Absolute Orientation IMU Fusion BNO055 Module [8]

3.3.2. Power System

The power architecture is designed around a primary battery that provides supply to the Raspberry Pi 5 and the charging circuitry for wireless subsystems. Secondary batteries are allocated to support local operation of subsystem sensors and microcontrollers. The output of the primary battery is regulated to a safe operating voltage for the Raspberry Pi 5 through a step-down converter. Battery state-of-charge is monitored by the Raspberry Pico 2W, which transmits the data via Bluetooth to the Raspberry Pi 5. This allows the user to receive real-time information on remaining battery life and anticipate recharge requirements.

The main battery should supply sufficient voltage which will then be stepped down to provide a stable voltage to the Raspberry Pi 5 as well as providing a high battery capacity for minimal downtime due to charging. Weight is also a major factor in choosing the main battery since the R.E.P. Tracker's final maximum weight requirement is set to 22.05 lb. Table 3-4 shows the options considered as the main battery.

Table 3-4: Processor/Charging System Battery Options

Main Battery	**Battery Voltage**	**Battery Capacity**	**Weight**	**Price**
Requirements	>5V	>10Ah	<5 lbs.	<$50
Eryy 12.8V 15Ah LiFePO4 Battery [10]	12.8 V	15 Ah	3.4 lbs.	$36.09
12V 18 Ah Uplus New Technology Battery [11]	12 V	18 Ah	11.8 lbs.	$37.99
KBT 12v 12ah Rechargeable Lithium-Ion Battery [12]	12 V	12Ah	2.4 lbs.	$42.99

Figure 3-5 shows the battery chosen as the processor and charging system battery. This battery was chosen because of its lower weight but high battery capacity.

Figure 3-5: Eryy 12.8V 15Ah LiFePO4 Battery [5]

The submodule batteries that are responsible for powering wireless subsystems need to be compact yet have a good battery life with sufficient voltage. Table 3-5 shows the options considered for the secondary battery below.

Table 3-5: Submodule Battery Options

Secondary Battery	Battery Voltage	Battery Capacity	Overall Size (LxWxH)	Price
Requirements	>3V	>=2000 mAh	<2.5 in x2 in x 0.5 in	<$50
Hiteuoms 3.7 V 2000 mAh LiPo Battery [13]	3.7 V	2000 mAh	2.09 in x 1.34 in x 0.39 in	$18.00 (4 pack)
MakerFocus 3.7 V 2000 mAh LiPo Battery [14]	3.7 V	2000 mAh	1.96 in x 1.3 in x 0.39 in	$9.99

Figure 3-6 shows the battery chosen as the submodule battery. This battery was chosen because of its lower cost while maintaining a compact size and sufficient battery capacity.

Figure. 3-6: Hiteuoms 3.7 V 2000 mAh LiPo Battery [6]

The power sensor is responsible for monitoring the wireless subsystem battery power/life. It should be low profile and deliver accurate data pertaining to the battery. Table 3-6 shows the options considered as the power sensor.

Table 3-6: Power Sensor Options

Power Sensor	**Measurement Capability**	**Microcontroller Connectivity**	**Weight**	**Thickness**	**Price**
Requirements	Voltage and Current	Able to read data with Pi 5	<2 oz	<0.25 in	<$10
Adafruit Ina260 High or Low Side Voltage, Current, Power Sensor [15]	Yes	Yes	0.1oz	0.1 in	$9.95
INA219 High Side DC Current Sensor [16]	Yes	Yes	0.1 oz	0.1 in	$9.95

Figure 3-7 shows the INA260 chosen to monitor the battery life of the unit. Both sensors considered fit the requirements, but the INA260 was ultimately chosen due to its wider range of sensing and flexibility.

Figure 3-7: Adafruit INA260 High or Low Side Voltage, Current, Power Sensor [15]

The charging controller is responsible for controlling secondary/subsystem batteries. It should be able to prevent overcharging and switch automatically between charging and normal discharge. Table 3-7 shows the options considered as the submodule battery charging controller.

Table 3-7: Charging Controller Options

Charging Controller	Charging Safety	Thickness	Microcontroller Connectivity	Price
Requirements	Prevent Over-Charging	<0.5 in	Can read statuses	<$10
Adafruit BQ25185 USB / DC / Solar Lithium Ion/Polymer Charger [17]	Yes	Yes	Yes	$6.95
Adafruit BQ25185 USB / DC / Solar Charger with 5V Boost Board [18]	Yes	Yes	No	$8.95

Figure 3-8 shows the BQ25185 without a 5V boost board chosen to charge the secondary/subsystem batteries. It was chosen because it could be easily connected to the microcontroller to read charging status.

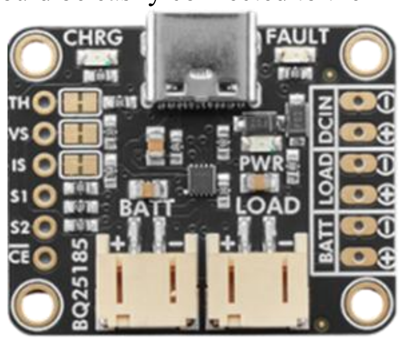

Figure 3-8: Adafruit BQ25185 USB / DC / Solar Lithium Ion/Polymer Charger [17]

The step-down converter is responsible for converting the 12.8 V main battery source to a safe operating voltage. This is essential to ensure the Raspberry Pi 5 is not receiving more voltage than it is rated for. Table 3-8 shows the step-down converter options.

Table 3-8: Step Down Converter Options

Step Down Converter	**Conversion Range**	**Implementation**	**Connection Type**	**Price`**
Requirements	12V to 5 V	Simple	Variety	<$10
Elecbee 24 V / 12 V To 5 V 5 A DC-DC Step Down Converter [19]	Yes	Yes	Yes	$4.99
Ekylin DC 12v 24v to 5v Step Down Converter Regulator 5A 25W [20]	Yes	Yes	No	$12.99

Figure 3-9 shows the step-down converter chosen for implementation. It offers a variety of connection types for input and output. Input offers a barrel jack connector and a breakout for hardwiring. The output offers a USB connection as well as a hardwire breakout. This was the only option considered due to its versatility and meeting all stated requirements.

Figure 3-9: Elecbee 24 V / 12 V To 5 V 5 A DC-DC Step Down Converter [19]

Figure 3-10 shows an overview of the power system's code.

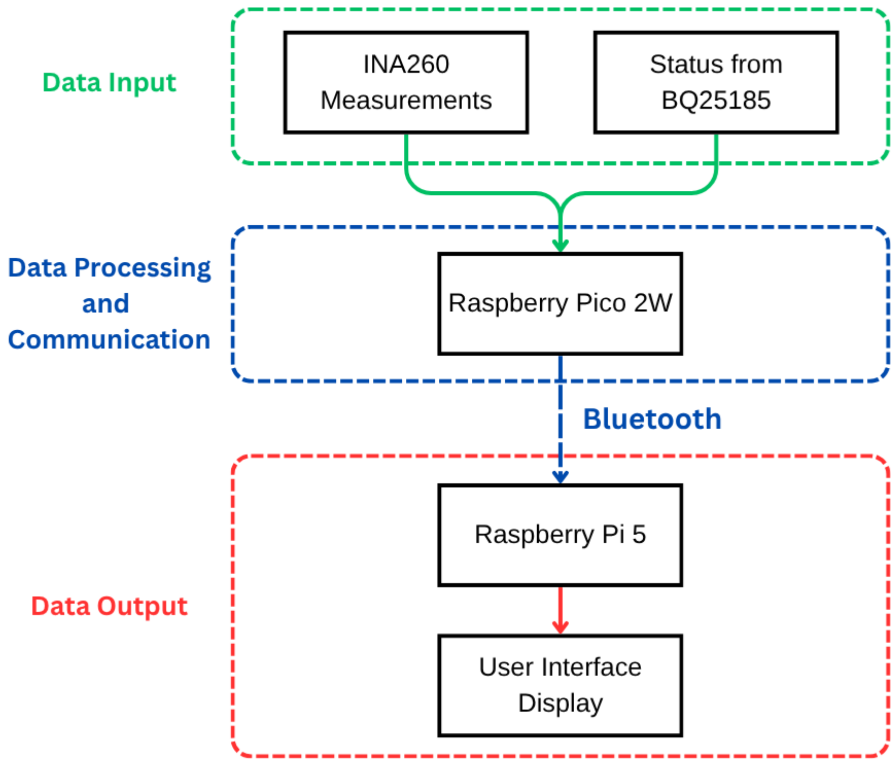

Figure 3-10: Power System Code Flowchart

The power system flowchart shown in Figure 3-10 shows the process of sensors taking in data. This data is read into the Raspberry Pico 2W then sent over Bluetooth to the main processor for necessary calculations and displayed on the user interface.

3.3.3. User Interface

The User Interface serves as the main display and means of user interaction. The touchscreen allows for ease of use and navigation without the need for additional controls. An exerciser can create a profile, load saved profiles with data, and train with the provided workout recommendations. All the carefully calculated suggestions for a workout routine are tailored for that specific user profile. After training starts, the user can view their current rep, set, and rest time. Table 3-9 below contains the parts options of the touchscreen display for the UI.

Table 3-9: UI Display Screen Options

Screen	Touchscreen	Resolution	Refresh Rate	Price
Requirements	**yes**	**>740p**	**at least 60Hz**	<$50
Hosyond 800x480 Display[21]	yes	800x480	60Hz	$39.99
Hosyond 1024x600 Display[22]	yes	1024x600	60Hz	$46.99

Figure 3-11 shows the chosen screen; the screen was chosen partly because it provides a clearer display. This display works better for displaying the camera that captures 1080p. A user can navigate the device via a full HD touchscreen [22].

Figure 3-11: 7" touchscreen 1024x600 HD display from Hosyond [22].

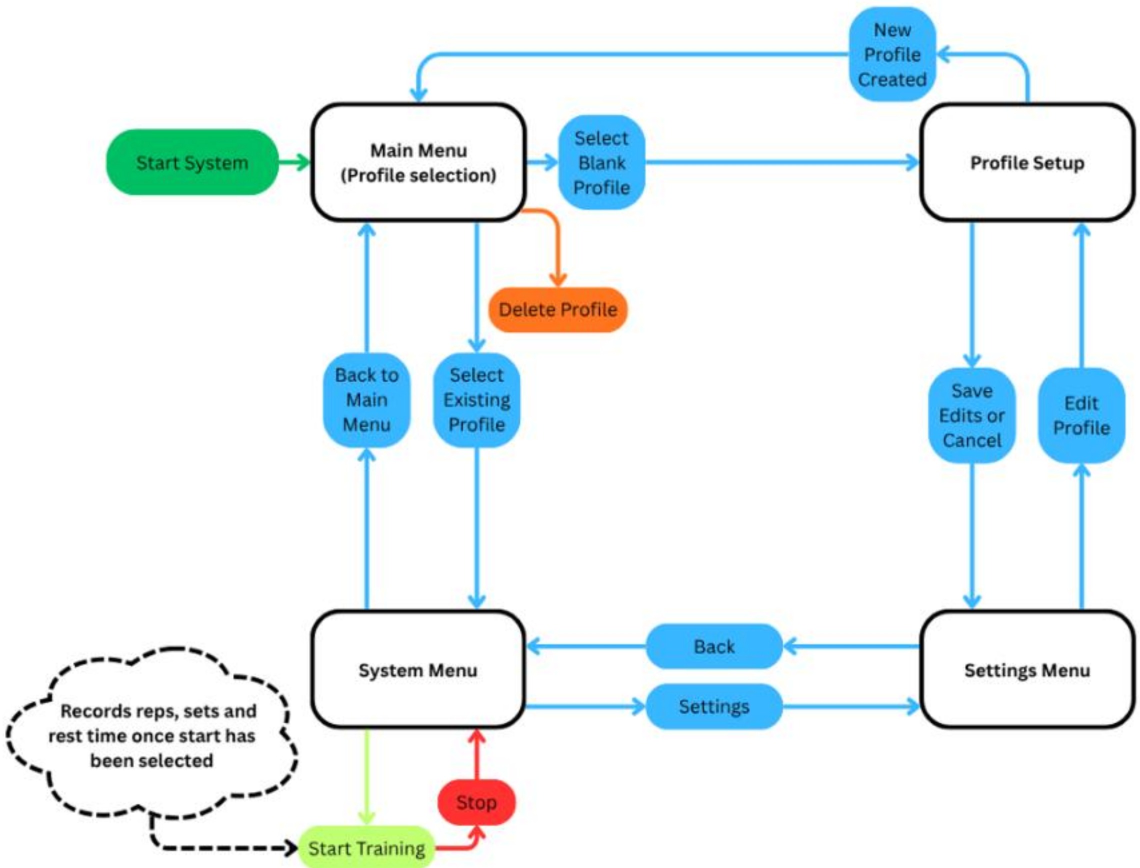

Figure 3-12: UI Navigation

The UI can be navigated using the touchscreen. A user can select or create a profile, train with professional recommendations, and edit or customize various preferences in settings.

3.3.4. Camera Analysis

The camera analysis enables form monitoring during weightlifting, providing visual data for real-time analysis and error detection. The feed integrates with the Raspberry Pi 5 through a USB connection, utilizing V4L2 drivers and OpenCV for video capture and processing. When a lifting fault occurs, such as excessive bar tilt, the system records and flags the corresponding footage. The error location is displayed and highlighted to facilitate recognition, evaluation, and correction. Table 3-12 shows the Anker PowerConf C200 2k Webcam as the final choice.

Table 3-10: Camera Options and Specifications

Product	Frame-Rate	Sensor	Focus	Connectivity	Resolution
Requirements	30 fps	≥2MP	Autofocus	N/A	≥1080p
TOALLIN 1080p HD Wireless Webcam [23]	30 fps	2MP	No	Wireless (USB receiver)	1080p
Anker PowerConf C200 2k Webcam [24]	30 fps	5MP	Yes	Wired USB	1080p

The **Anker PowerConf C200 2K Webcam,** which is shown in **Figure 3-13**, was selected for its implementation simplicity and reduced weight. The initial configuration employed a wireless webcam mounted on a tripod to allow flexible positioning. The revised design incorporated a wired webcam mounted directly to the case, eliminating the need for wireless capability and external mounting hardware. This modification decreased system weight and complexity. Latency remained minimal, measured below 50 ms for the wired USB connection compared to 100–200 ms for wireless operation.

Figure 3-13: Anker PowerConf C200 2k Webcam [24]

In conclusion, the selection of the Anker PowerConf C200 2K Webcam provides a balanced combination of performance, reliability, and integration simplicity. Its wired USB connectivity ensures low-latency operation essential for real-time form analysis, while its higher-resolution sensor and autofocus capabilities enhance detection accuracy during lifts. By reducing system weight, eliminating unnecessary wireless components, and improving processing responsiveness, the final camera configuration effectively supports the overall goal of delivering consistent and accurate movement assessment within the weightlifting monitoring system.

3.3.5. Performance Monitoring

The main components of the Performance Monitoring subsystem are a pulse oximeter and temperature sensor to measure and track the user's heart rate, blood oxygen level, and skin temperature data. These components will be on a wristband that measures this data. This data is then given to the Pico 2W and transmitted to the Raspberry Pi 5 via Bluetooth. The error range for the measured values should be less than 10% for Heartrate and Temperature, but less than 3% error for Blood Oxygen [25]**.** As shown in the tables below, there are several oximeters and temperature sensors to choose from for this project.

The pulse oximeter is responsible for measuring the heart rate and blood oxygen of the user. It should be able to emit the correct light wavelengths to track these metrics accurately so that they are within the previously mentioned error ranges. Table 3-11 shows some available options for the pulse oximeter.

Table 3-11: Pulse Oximeter Options

Product	**Operating Voltage (V DC)**	**LED Wavelength (nm)**	**Size(mm)**	**Cost**
Requirements	3.3 - 5 V DC	RED 660 - 700 nm, IR 860 - 900 nm	5 – 6 × 3 – 5 × 1 – 2 mm	< $15
AFE4490 (TI) Integrated Analog Front End Oximeter [26]	3.0 – 5.25 V DC	Red 660 nm, IR 940 nm	6 × 6 mm	$149.0
MAX30102 Pulse Oximeter [27]	3.3 - 5 V DC	Red 660 nm, IR 880 nm	5.6 × 3.3 × 1.55 mm	$5.41
BH1792GLC Biometric Sensor [28]	2.5 – 3.6 V DC	Green: 525 nm	2.8 × 2.8 × 1.0 mm	$11.49

Figure 3-14 shows the MAX30102 which was chosen because it meets the specified requirements. It provides the correct wavelength at an affordable cost with a compact size.

Figure 3-14: **MAX30102 Pulse Oximeter and Heart Rate Sensor [27]**

The temperature sensor is responsible for measuring the skin temperature of the user. It should be a compact size that does not hinder the user in any way. It should also be capable of measuring the normal temperature range of the human body. Table 3-12 shows a few available options.

Table 3-12: Temperature Sensor Options

Product	**Operating Voltage (V DC)**	**Temperature Range**	**Size**	**Cost**
Requirements	3 – 5 V DC	0 - 50 °C	8 - 10 × 10 - 12 × 1 - 2 mm	< $15
LilyPad Temperature Sensor [29]	2.7 – 5 V DC	0 - 70 °C	20 mm Ø × 0.8 mm thick	$5.95
DFRobot TS01 Non-Contact IR Sensor [30]	5 – 24 V DC	–70 - 380 °C	15.4 mm Ø × 78 mm length	$89
MLX90632 Infrared Temperature Sensor [31]	3.0 – 3.6 V DC	-20 - 100 °C	3 × 3 × 1 mm	$14.90

Figure 3-15 shows the chosen temperature sensor. This sensor was chosen for its small size as well as its simplicity of function. It uses a thermistor to calculate temperature based on voltage and can merely rest against the user's skin to achieve this.

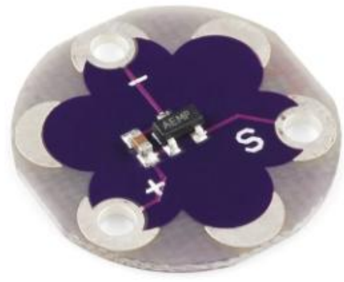

Figure 3-15: Lilypad Temperature Sensor [29]

Overall, the MAX30102 was the best choice budget-wise for an oximeter that emits the correct light wavelength at an acceptable module size. The Lilypad Temperature Sensor is an excellent choice for measuring skin temperature as it is discreet and does its job effectively. These two modules are the most fitting for the intended purpose of tracking the user's health data.

3.3.6. User Feedback

The User Feedback subsystem centers around collecting data from multiple subsystems through the Pi 5 and processing it using dedicated feedback functions. User Feedback begins with User inputs received through the User Interface (UI), which determines parameters such as audio levels, Bluetooth or speaker outputs, and mute preference.

The main stage of the process lies in real-time feedback, where data from various subsystems is continuously analyzed. The Motion Tracking data serves as the most critical input, evaluated by feedback functions to classify the user's form as incorrect, inadequate, or correct. While this data is being processed, ongoing system checks ensure that any faults or emergency conditions are promptly detected and addressed.

Finally, the processed feedback is delivered through three distinct output modes:

1. Audio output – an optional channel that can be muted by the user.
2. Data logging – a non-optional process that continuously stores performance and feedback logs within the Pi 5, accessible through the UI
3. LED indicators – a visual feedback system where LED's display red, yellow, or green to represent incorrect, inadequate, and correct form respectively

Figure 3-16 shows the flow of information in the user feedback and how it outputs**.**

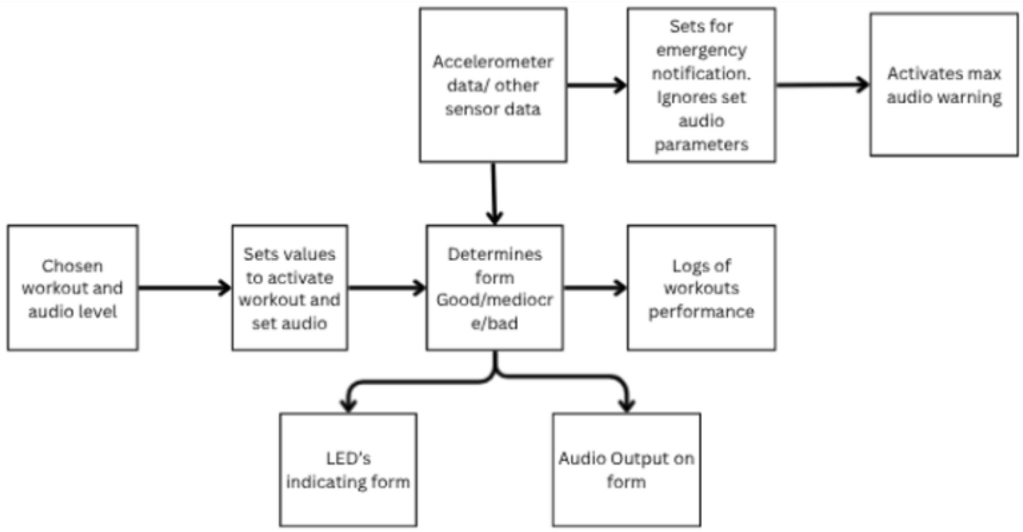

Figure 3-16 User Feedback's Input/Output Diagram

Overall, the User Feedback system collects data from multiple sensors and processes it either continuously to provide real-time guidance or periodically to detect conditions that require emergency feedback. Figure 3-16 outlines how specific conditions move through the feedback flow, with emergency pathways operating independently of user-selected audio settings, for example.

3.4. Level 2 Prototype Design

In Design II, the full prototype of the R.E.P. Tracker aims to complete the integration of all subsystems, including the currently partial Performance Monitoring, Camera Feed, and User Feedback, into a fully functional, cordless device that delivers seamless real-time coaching for weightlifters. This phase will focus on refining wireless communication for low-latency data transfer, optimizing battery efficiency to extend usage beyond initial prototypes, and incorporating advanced software algorithms for more precise form analysis and vital monitoring. By addressing any remaining compatibility issues between the Raspberry Pi 5 central microcontroller and peripheral Pico 2W units, the prototype will achieve enhanced portability in a compact tote case, ensuring compliance with FCC standards while maintaining accuracy in motion tracking and vital readings. Ultimately, this will result in a user-friendly system that provides comprehensive, unsupervised workout guidance through visual, audio, and logged feedback, paving the way for final testing and potential commercialization.

3.4.1 Level 2 Diagram

Figure 3-17 shows the final prototype design for the R.E.P. Tracker. The functionality depends on the hardware components working together. The display is a Hosyond 7" Raspberry Pi 5 touchscreen, connected to the Raspberry Pi 5, which acts as the central microcontroller. The touchscreen connects through a micro-HDMI to micro-HDMI cable, while communication with other sensors occurs wirelessly through a Raspberry Pi Pico 2W using Bluetooth. The Motion Tracking system uses a BNO055 gyroscope/accelerometer coded on the Pico 2W, which transmits data to the Pi 5 via Bluetooth. Performance Monitoring utilizes a Lilypad skin temperature sensor and a MAX30102 heart rate/O_2 sensor, also coded on the Pico 2W and sending data wirelessly to the Pi 5. The Power System uses a 3.7V LiFePO4 rechargeable battery with a BQ25185 battery charger and an INA260 power sensor to manage power delivery and charging for the sensor modules. These modules are charged with a 12V LiFePO4 battery, which also powers the Pi 5 through an Elecbee 12V to 5V 5A DC step-down converter. The Audio Output is handled either through the built-in speakers on the back of the touchscreen or via Bluetooth, providing flexible and convenient sound feedback options.

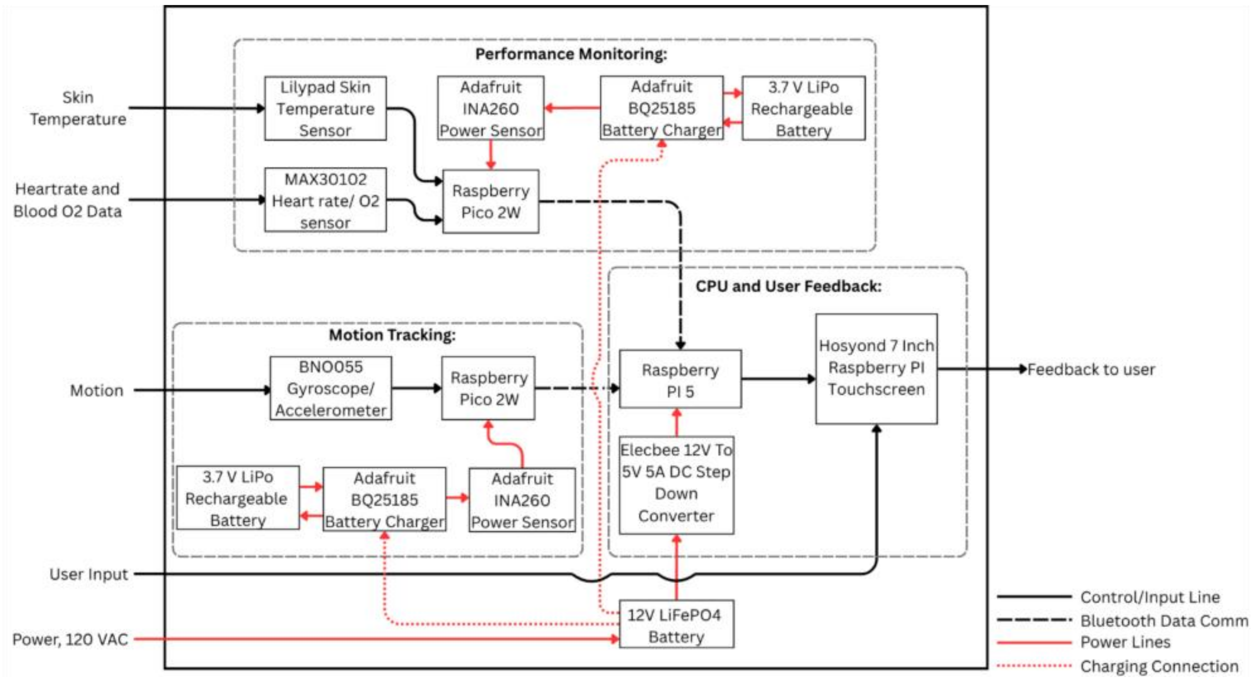

Figure 3-17: Diagram for The R.E.P. Tracker (Level 2).

The previous sections have thoroughly outlined the design and approach behind the R.E.P. Tracker. For the system to operate effectively, all subsystems must work in harmony to provide consistent and reliable assistance. Achieving this requires that each subsystem not only functions independently but also integrate seamlessly with the others, avoiding interference or performance conflicts. The following sections detail the testing procedures and results for each subsystem to ensure optimal coordination and overall system functionality.

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